Thermal resistance of interfaces in hybrid bonding process

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***Abstract*—Thermal interfaces of both oxide and metal in hybrid bonding limit heat removal in three-dimensional integrated circuits (3D ICs). This summary focus on thermal resistance of SiOx-SiOx and Cu-Cu/dielectric interfaces in hybrid bonding process via advanced thermal characterization, highlighting the significance of high-quality bonding.**

I. Introduction And Background

With the evolution of integrated circuits toward three dimensions, heat dissipation has become a critical bottleneck. In 3D-ICs, hybrid-bonding interfaces are key thermal pathways. However, the thermal resistances of the bonded interfaces are much less studied than the correponding electrical resistanes[1]. This work reports the experimental investigation of the thermal resistance of low-temperature bonded SiO_x-SiO_x interfaces and hybrid Cu-Cu/dielectric interfaces via time-domain thermoreflectance (TDTR) technique. Unlike the previous studies on the thermal resistance of Cu-Cu bonding interfaces [2], we also study on the thermal resistance of bonding interfaces in real 3D-IC chips, revealing how interface quality contributes to thermal resistances.

II. Results

SiO_x–SiO_x specimens were fabricated by thermally growing ≈500 nm amorphous SiO_x on Si wafers, performed surface plasma-activation, and bonded at 250°C for 2 h [3]. Samples were CMP-polished to expose a stair cross section (upper SiO_x ≈100 nm) and coated with ≈80 nm Al transducer for TDTR measurement (Fig. 1(a)). Picosecond-acoustics echoes in the in-phase signal were used to locate the Al–SiO_x interface, the SiO_x–SiO_x interface, and the SiO_x-Si interfaces. Measurement spots are selected with the desired upper-oxide thickness (Fig. 1(b)), enabling TDTR lasers to penetrate through the buried bonded interface.

As a result, comparative TDTR ratio curves (Fig. 2) recorded at different modulation frequencies and spot sizes show virtually indistinguishable responses for regions with and without the SiO_x–SiO_x interface. Consequently, extremely low interfacial thermal resistance is observed. Fitting data indicate the overall SiO_x thermal conductivity (κ)≈1.36 $W \cdot m^{-1} \cdot K^{-1}$; simultaneous fitting sets a lower-limit for SiO_x–SiO_x thermal boundary conductivity (TBC) at ≈150 $MW \cdot m^{-2} \cdot K^{-1}$, equivalent to the thermal resistance of a dielectric layer <~9.2 nm. Further atomic-scale and compositional characterizations confirm an indistinguishable void-free oxide–oxide interface. Therefore, for high-quality low-temperature hybrid bonds, the SiO_x–SiO_x thermal resistance is negligible.

Additionally, a 3D-IC chip with 2-µm-pitch hybrid bonding was also researched to assess real-chip thermal behavior [4]. By removing the top die, the bonded Cu pads on the bottom die were exposed. TDTR characterization on both electrically connected and unconnected regions indicated an effective thermal conductivity of the bonding layer of ≈9–12 $W \cdot m^{-1} \cdot K^{-1}$. This result suggests that the average effective thermal conductivity of the metal pads and the surrounding dielectric exhibit relatively low thermal resistance at the hybrid bonding interfaces, attributed to advanced bonding processes that yield high-quality interfaces.

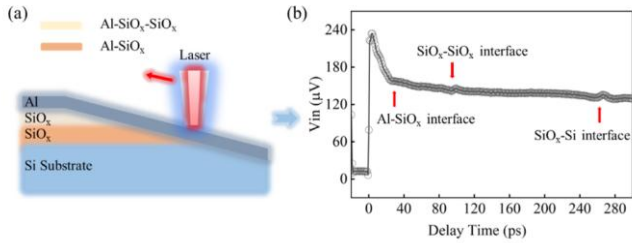


Fig. 1. The schematic diagram of the polished sample (a) and the echoes of different interfaces (b) (reproduced with permission from [3]).

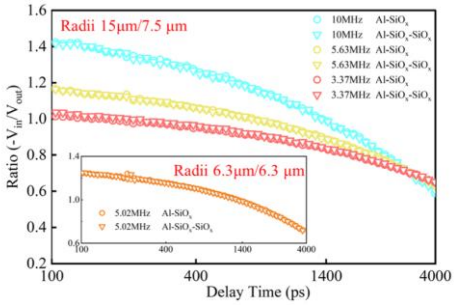


Fig. 2. Comparison of TDTR ratio data between Al-SiOx sample structure and Al-SiOx-SiOx sample structure with different modulation frequencies and pump/probe radii (reproduced with permission from [3]).

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References

[1] Y. Huang *et al.*, "Cu-Based Thermocompression Bonding and Cu/Dielectric Hybrid Bonding for Three-Dimensional Integrated Circuits (3D ICs) Application," *Nanomaterials*, vol. 13, no. 17, 2023, pp. 2490.
[2] H. Kim *et al.*, "Thermal transport properties of hybrid bonding with passivation," in *Proc. 2024 IEEE 74th Electronic Components and Technology Conference (ECTC)*, 2024, pp. 1609-1612.
[3] X. Zhang, L. Chang, L. Li, and Z. Cheng, "Thermal conductance across bonded SiO_x–SiO_x interfaces in hybrid bonding process," arXiv:2601.03106, 2026.
[4] X. Guo *et al.*, "First Demonstration of 3D Thermal Characterizations and Database in TSV-Embedded 3D-IC Stacks with >9 W/m-K and 2-um-pitch Hybrid Bonding," submitted to *IEEE VLSI*, 2026.

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